

CYBERSECURITY 60-DAY CHALLENGE

Task 2 — Network Security & Scanning

Days 13–24 | Practical Lab Report

Objective

Learn reconnaissance, network scanning, vulnerability scanning, packet analysis and basic firewall controls in an isolated cybersecurity training environment.

Lab Scope and Target

Item	Value
Testing machine	Kali Linux
Target	Metasploitable2
Target IP	192.168.56.102
Network	VirtualBox Host-Only isolated lab
Scanner	OpenVAS / OpenVAS Default

Nmap Service Surface

The verified lab service scan identified 23 open TCP services on Metasploitable2.

Port	Service
21/tcp	FTP
22/tcp	SSH
23/tcp	Telnet
25/tcp	SMTP
53/tcp	DNS
80/tcp	HTTP
111/tcp	RPCbind
139/tcp	NetBIOS-SSN
445/tcp	Microsoft-DS/SMB
512/tcp	exec
513/tcp	login
514/tcp	shell
1099/tcp	Java RMI
1524/tcp	Ingreslock
2049/tcp	NFS
2121/tcp	FTP proxy
3306/tcp	MySQL
5432/tcp	PostgreSQL
5900/tcp	VNC
6000/tcp	X11
6667/tcp	IRC
8009/tcp	AJP13

Port	Service
8180/tcp	HTTP/unknown

The exposed surface includes legacy protocols and multiple application/database services. In production environments, unnecessary services should be disabled, patched and access-restricted.

OpenVAS Vulnerability Assessment

Metric	Observed result
Hosts scanned	1
Ports	20 of 23
Applications	20 of 20
CVEs	36 of 36
Results	69 of 635
TLS certificates	2 of 2
Critical	13
High	10
Medium	40
Low	6
Highest reported CVSS	10.0 (Critical)
Scan duration	1:11 h

The severity counts total 69 findings. The supplied screenshots do not show the complete individual vulnerability list; therefore no unsupported CVE names or remediation commands are claimed here.

OpenVAS Evidence

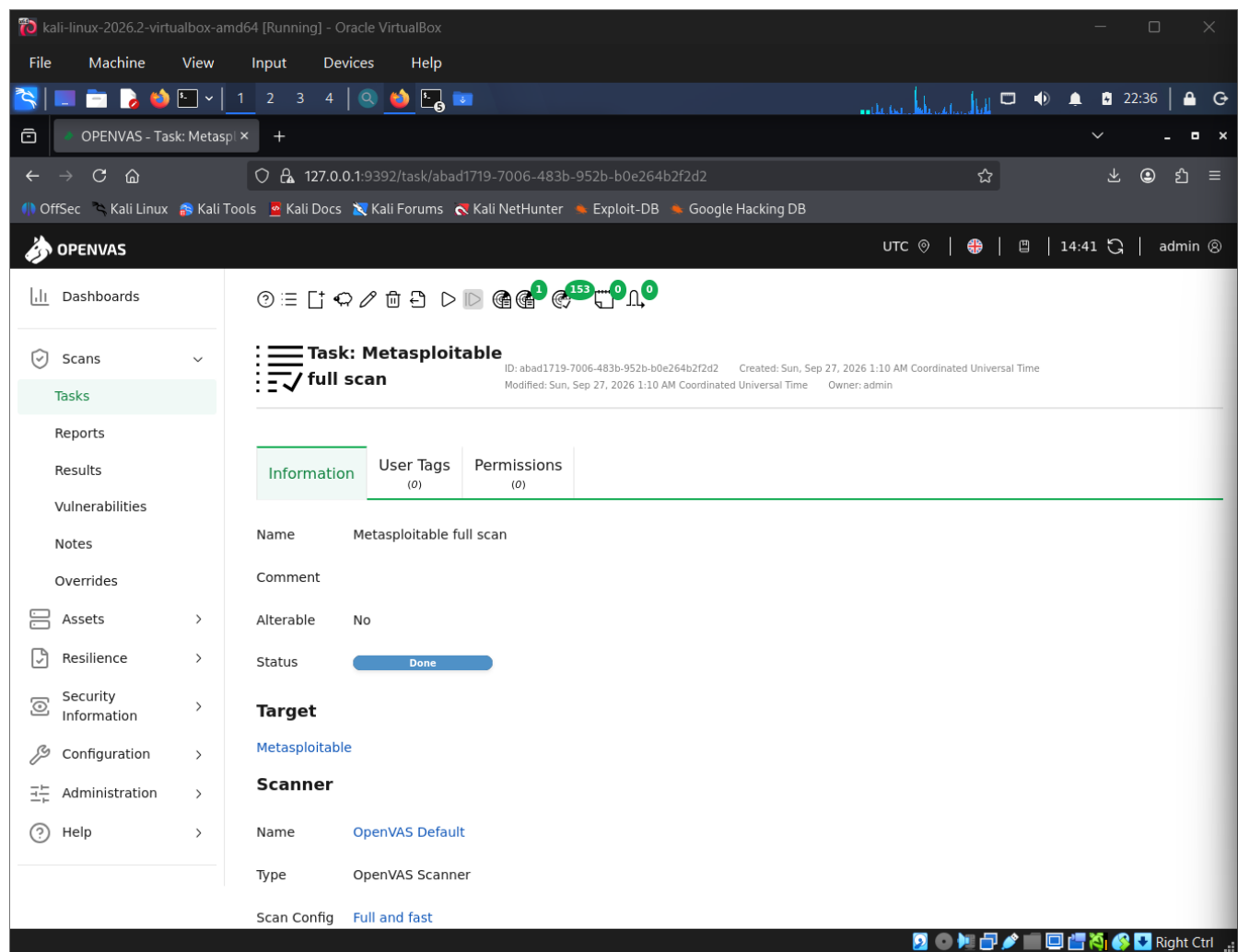


Figure 1. Completed OpenVAS task.

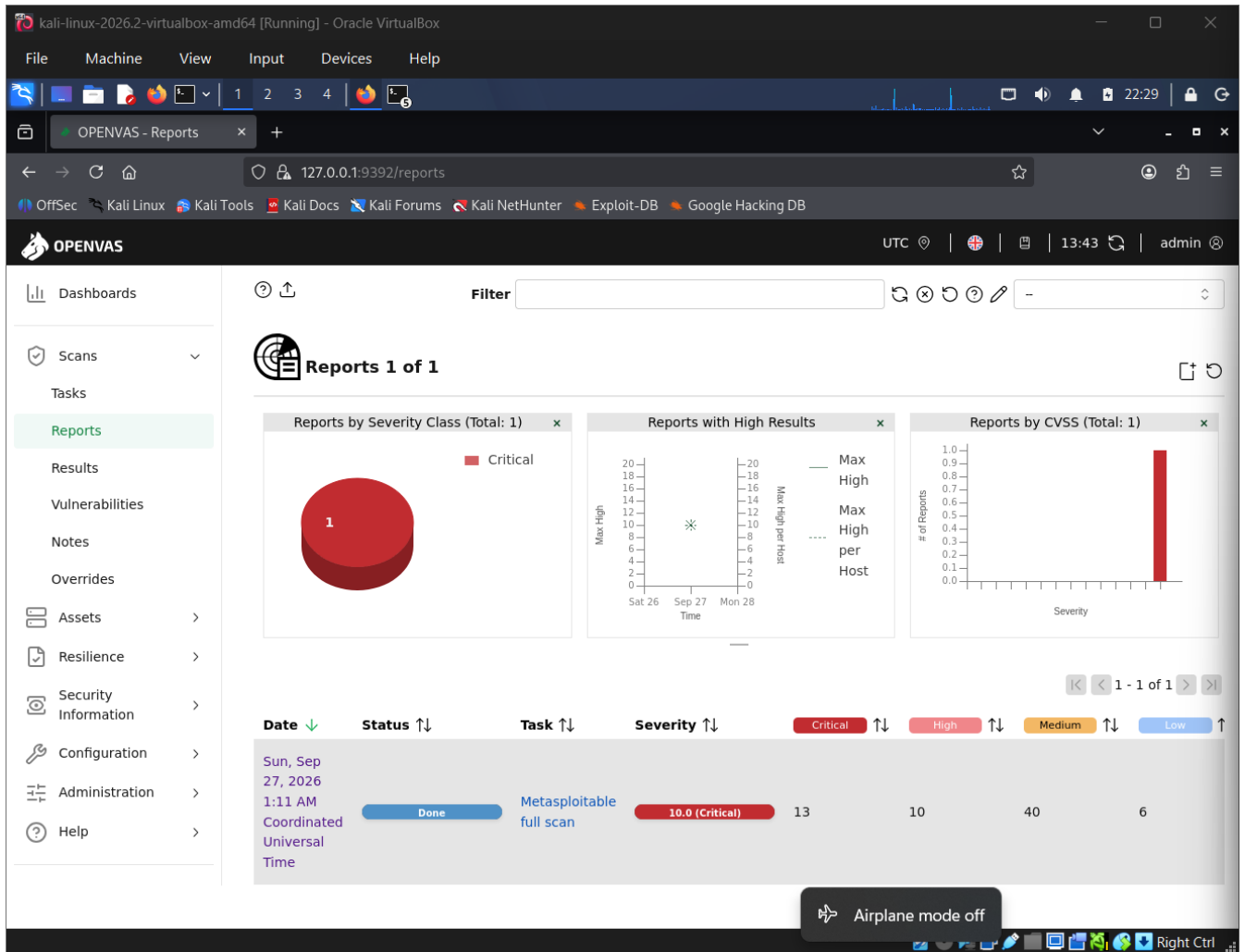


Figure 2. OpenVAS report summary.

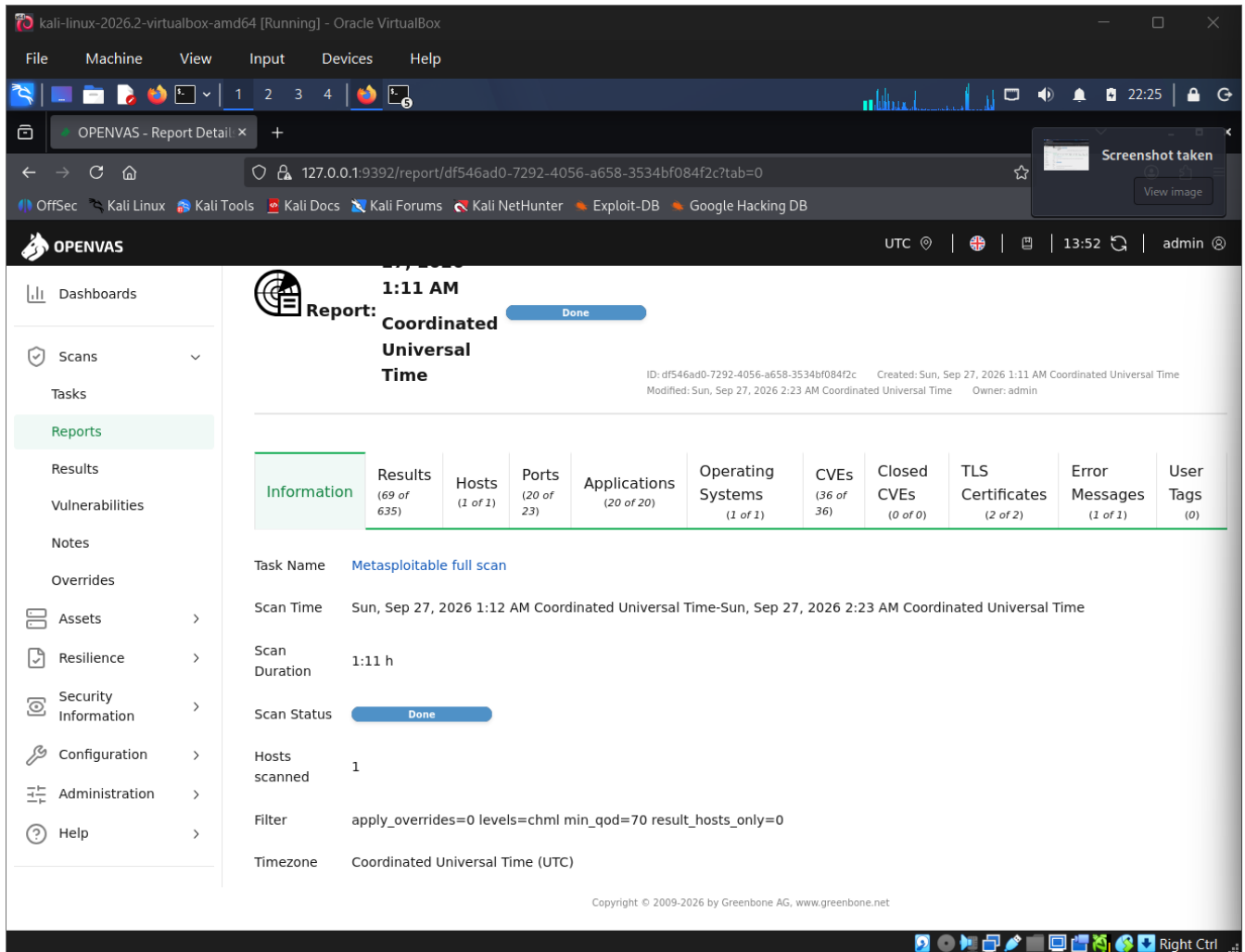


Figure 3. OpenVAS report details.

Findings and Defensive Interpretation

- 13 Critical, 10 High, 40 Medium and 6 Low results were shown.
- The highest reported CVSS score was 10.0 (Critical).
- Nmap identifies exposed services; OpenVAS provides vulnerability assessment context.
- Real-world remediation would involve validation, patching, disabling unnecessary services, restricting access and re-scanning.
- The testing scope was the intentionally vulnerable Metasploitable2 VM in an isolated VirtualBox Host-Only lab.

Evidence note: the uploaded screenshots do not include every requested Nmap, Wireshark, reconnaissance and firewall screenshot. Add those images to the final GitHub evidence folder if required by the evaluator.